Sharath N Payyadi

Robotics & Simulation Engineer | ROS2 · NVIDIA Isaac Sim/Isaac Lab · AMR & Manipulation

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Nationality: Indian | Languages: English (Fluent), German (A2)

PROFILE

Robotics and Simulation Engineer with 3+ years of experience developing ROS1/ROS2 software for autonomous mobile robots and robotic arms, including two years of on-site production experience in Germany (BMW, NEOM). Currently focused on simulation-driven development using NVIDIA Isaac Sim and Isaac Lab for sim-to-real validation, imitation learning, and teleoperation. Combines strong software infrastructure skills (CI/CD, Docker, Git) with hands-on simulation and motion-planning expertise (MoveIt, Nav2).

PROFESSIONAL EXPERIENCE

Robotics Engineer Nov 2025 – Present

Fireloop AI

- Develop ROS2 modules for robotic arms and autonomous mobile robots (AMRs), with validation and testing performed in NVIDIA Isaac Sim prior to real-world deployment.
- Set up and configure customer environments to support field deployment of robotic systems.
- Work on imitation learning and teleoperation pipelines using NVIDIA Isaac Lab, covering environment setup, dataset collection, and model training.

Software Developer – Robotics & Infrastructure Automation Apr 2023 – Aug 2025

AITONOMI AG, Düsseldorf, Germany

- Built a Jenkins CI/CD pipeline that reduced Docker image build and packaging time from 2–3 days to 0.5 day, and cut deployment testing time from 2–3 days to 0.5 day.
- Supported on-site commissioning during Factory and Site Acceptance Tests (SAT) for two production deployments: a large AGV at BMW (Germany) and an autonomous truck at NEOM Port (Saudi Arabia), coordinating with a cross-functional team of ~20 engineers.
- Led migration of the software stack from ROS1 to ROS2, adopting DDS, QoS profiles, and lifecycle node management to improve system reliability.
- Developed a monitoring module for production robot fleets to support real-time operational visibility.

Master's Thesis – Traffic Management of Multiple Robots in a Graph Environment Apr 2022 – Dec 2022

NODE Robotics GmbH, Stuttgart, Germany

- Designed a centralized traffic management system using a topological graph representation (sections, corridors, crossings, POIs) to coordinate heterogeneous AMR fleets (MiR100, Geek+) communicating via the VDA5050 protocol over MQTT.
- Built a conflict detector, dynamic detour generator, and safe-spot calculator to resolve path conflicts proactively, achieving zero deadlocks across all validated multi-robot scenarios, including corridor crossings and 3-robot cascading conflicts.
- Validated the system in RViz simulation and on physical MiR100/Geek+ robots; implemented in ROS2 (C++) with a Google Test suite integrated into the CI/CD pipeline, reducing robot downtime in shared spaces by 10%.

Student Intern – ROS Backend Developer Sep 2021 – Feb 2022

NODE Robotics GmbH, Stuttgart, Germany

- Developed backend functionality for a ROSBAG Player UI tool enabling real-time playback and error analysis for production-deployed robots.
- Integrated WebSocket and REST APIs to improve system communication and data-handling performance.

PROJECTS

Pick and Place using TIAGo Robot Apr 2021 – Oct 2021

Hochschule Darmstadt

- Implemented ArUco marker-based object detection and MoveIt motion planning for a mobile manipulator pick-and-place task; compared real-world vs. Gazebo simulation performance.

Disease Detection in Paddy Crop using CNN Algorithm Jun 2019 – Jun 2020

Visvesvaraya Technological University (VTU), Bengaluru, India

- Designed and implemented a deep learning-based approach using a CNN algorithm with a trained Keras model and OpenCV to detect plant diseases at an early stage.
- Published findings in the International Journal of Recent Technology and Engineering (IJRTE), Vol. 8(6), March 2020.

EDUCATION

M.Sc. in Electrical and Information Technology Sep 2020 – Dec 2022

Hochschule Darmstadt, Germany

B.E. in Electronics and Communication Engineering Jul 2016 – Jun 2020

Visvesvaraya Technological University (VTU), Bengaluru, India

PUBLICATIONS & TECHNICAL WRITING

- "Disease Detection in Paddy Crop using CNN Algorithm", International Journal of Recent Technology and Engineering (IJRTE), Vol. 8(6), March 2020.
- Author of a technical blog on the NVIDIA robotics simulation ecosystem (Isaac Sim, Isaac Lab, Cosmos, GR00T) — read on Medium.

TECHNICAL SKILLS

- Simulation: NVIDIA Isaac Sim, NVIDIA Isaac Lab, Gazebo, Sim-to-Real Workflows, Imitation Learning, Teleoperation
- Robotics Middleware: ROS1, ROS2 (DDS, QoS, lifecycle management), MoveIt, MoveIt2, Nav2
- Programming: C++ (17+), Python, Bash
- AI/LLM Tooling: Gemini Vision, Gemini, FastAPI, Agentic Pipelines
- DevOps: Jenkins, Git, Bitbucket, Docker, CI/CD Pipelines
- Sensors & Perception: LiDAR, Cameras, IMU, Object Detection
- OS & Tools: Linux, Windows, Atlassian Suite (Jira, Confluence)

EXTRACURRICULAR

Volunteered in Hochschule Darmstadt's Buddy Program supporting incoming international students, and organized internal workshops and pair-programming sessions to improve team tech skills.